F. Omega Numbers

time limit per test: 3 seconds

memory limit per test: 512 megabytes

For a given number n, consider the function $\omega(n)$, which is equal to the number of unique prime numbers in the prime factorization of the number n.

For example, $\omega(12) = \omega(2^2 \cdot 3) = 2$. And $\omega(120) = \omega(2^3 \cdot 3 \cdot 5) = 3$.

For an array of natural numbers a and a natural number k, we define $\mathrm{f}(a, k) = \sum_{i<j} \omega(a_i \cdot a_j)^k$ for all $i < j$.

You are given an array of natural numbers a of length n and a natural number k. Calculate $\mathrm{f}(a, k)$ modulo $998\,244\,353$.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains 2 integers n and k ($1 \le n \le 2 \cdot 10^5, 1 \le k \le 10^9$) — the length of the array a and the exponent of the operation, respectively.

The second line of each test case contains n natural numbers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le n$) — the array a.

It is guaranteed that the sum of n across all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single line containing an integer — the value of the function $\mathrm{f}(a, k)$ modulo $998\,244\,353$.

Example

input Copy

```
3
4 1
3 3 3 3
4 1
1 1 1 1
4 2
1 2 3 4
```

output Copy

```
6
0
12
```

Note

Explanation of the first test case example:

For any pair (i, j), the value $\omega(x)$ for the product is $\omega(3^2) = 1$. There are a total of 6 pairs, and the exponent is 1. The final answer is 6.

Explanation of the second test case example:

In any pair of the second test case, the product of the numbers in the pair equals 1, so the number of primes in it is 0. Therefore, the final answer is also 0.

Explanation of the third test case example:

Consider all pairs (i, j):

- $(1, 2)$: the product of the numbers at positions 1 and 2 equals $a_1 \cdot a_2 = 1 \cdot 2$, $\omega(2) = 1$.
- $(1, 3)$: the product of the numbers at positions 1 and 3 equals $a_1 \cdot a_3 = 1 \cdot 3$, $\omega(3) = 1$.
- $(1, 4)$: the product of the numbers at positions 1 and 4 equals $a_1 \cdot a_4 = 1 \cdot 4$, $\omega(2^2) = 1$.
- $(2, 3)$: the product of the numbers at positions 2 and 3 equals $a_2 \cdot a_3 = 2 \cdot 3$, $\omega(2 \cdot 3) = 2$.
- $(2, 4)$: the product of the numbers at positions 2 and 4 equals $a_2 \cdot a_4 = 2 \cdot 4$, $\omega(2^3) = 1$.
- $(3, 4)$: the product of the numbers at positions 3 and 4 equals $a_3 \cdot a_4 = 3 \cdot 4$, $\omega(3 \cdot 2^2) = 2$.

In the answer, the values of $\omega(x)$ are raised to the power of 2, so $1^2 + 1^2 + 1^2 + 2^2 + 1^2 + 2^2 = 12$.

Codeforces Round 1070 (Div. 2)

Finished

Practice

→ Virtual participation

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Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
363345314	Feb/17/2026 02:50	Accepted
363345166	Feb/17/2026 02:45	Accepted
363345087	Feb/17/2026 02:42	Accepted
363344936	Feb/17/2026 02:38	Accepted
363344748	Feb/17/2026 02:31	Runtime error on test 7
363344708	Feb/17/2026 02:30	Runtime error on test 7
363344520	Feb/17/2026 02:24	Runtime error on test 7
363344495	Feb/17/2026 02:23	Time limit exceeded on test 2
363344402	Feb/17/2026 02:21	Time limit exceeded on test 2

→ Problem tags

bitmasks combinatorics dp math number theory *2400

No tag edit access

→ Contest materials